SURAJ KUMAR

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TECHNICAL SKILLS

Frontend Development:	JavaScript (ES6+), React.js, Redux Toolkit, Zustand, HTML5, CSS3, Tailwind CSS
Backend & APIs:	Node.js, Express.js, RESTful API Design, Authentication (JWT, OAuth), Postman, Swagger
Databases:	PostgreSQL, MongoDB, Mongoose, Prisma ORM, Redis (Caching & Distributed Locks), Elasticsearch
Architecture & Real-Time:	WebSockets, Socket.io, WebRTC (P2P), Event-Driven Architecture, Microservices, Kafka
Cloud, DevOps & Tools:	Git, GitHub, Docker, Docker Compose, CI/CD (GitHub Actions), Vercel, Render, Jira, Linux
CS Fundamentals:	Data Structures & Algorithms, Object-Oriented Programming (OOP), OS, DBMS, Computer Networks

EXPERIENCE

Software Engineer Intern **Feb 2026 – Jul 2026**

Earth Services | Patna, India

- Assisted the engineering team in developing and maintaining frontend views and backend modules for an internal **loan collections and borrower case-tracking web portal**.
- Built responsive dashboard tables, search forms, and filter components using **React.js** and **Tailwind CSS** to help support staff view assigned borrower accounts and overdue delinquency statuses.
- Developed and updated **RESTful API** endpoints using **Node.js** and **Express.js** for borrower profile lookups, call-disposition remarks logging, and follow-up reminder updates.
- Created database schemas, basic CRUD queries, and server-side pagination using **MongoDB** and **Mongoose** to organize borrower account records and daily agent activity logs.
- Implemented login session handling and route-level access checks using **JSON Web Tokens (JWT)** middleware so team leads and agents could only view their assigned cases.
- Tested API endpoints, request payloads, and status codes using **Postman**, and collaborated with senior developers using **Git**, **GitHub**, and **Jira** for bug fixes and code reviews.

PROJECTS

TrainTravel – Microservices Railway System [GitHub] **React, Node.js, PostgreSQL, Redis, Kafka**

- Designed a production-grade microservices architecture utilizing the database-per-service pattern (Prisma/PostgreSQL) and an API Gateway to enforce JWT validation, circuit breaking, and route-level rate limiting.
- Orchestrated complex distributed transactions using the **Saga pattern**, coordinating booking, inventory, and payment microservices asynchronously via **Apache Kafka** message queues and Dead-Letter Queue (DLQ) fallbacks.
- Engineered a highly concurrent seat reservation system leveraging **Redis distributed locks** to prevent overbooking, alongside an Elasticsearch-backed search service achieving sub-second train and station lookups.

Swiss Bank [GitHub] [Live] **React, Node.js, PostgreSQL, Prisma, Express.js, JWT**

- Architected an enterprise-grade banking platform utilizing a **double-entry ledger engine**, ensuring immutable transaction history and precise running balance calculations, eliminating floating-point errors.
- Engineered atomic, ACID-compliant peer-to-peer transfers by implementing **pessimistic row-level database locking** (lowest account number first) to prevent race conditions and deadlocks during concurrent transfers.
- Implemented an advanced security layer using Express sessions with HTTP-only cookies, BCrypt password/PIN hashing, and **2FA (TOTP)** to protect user assets and virtual debit card controls.

NoShare – Decentralized P2P File Transfer [GitHub] [Live] **React, WebRTC, WebSockets**

- Engineered a serverless file-sharing platform utilizing **WebRTC data channels**, successfully eliminating intermediate server storage costs and reducing file transfer latency by **60%**.
- Architected an Express.js signaling server fortified with an in-memory IP rate limiter to throttle room handshakes and mitigate potential DDoS flood vectors across peer networks.

EDUCATION

Bachelor of Computer Applications (BCA) **Jun 2020 – Aug 2023**

Aryabhatta Knowledge University | Patna, India

CERTIFICATIONS

Full Stack Web Development Course • Credential Link